MO YUNXIANG

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EDUCATION

The Hong Kong University of Science and Technology Sep 2024 – Present
BSc, Triple Major in Computer Science, Mathematics & Electronic Engineering; Extended Major in Artificial Intelligence
GPA: 4.1 / 4.3
Honors:

- **University's Scholarship Scheme for Continuing Undergraduate Students** (Top 1%), 2024
- **S.S. Chern Class** – *Honor for top academic performance across all mathematics coursework.*
- **Dean's List Honor** (GPA above 3.7), 2024, 2025

Relevant Coursework: Algorithms, Statistical Inference, Machine Learning, Linear Algebra, Probability Theory *(A+ in all CS/Math courses)*

Beijing 101 High School Sep 2021 – Jun 2024
Gaokao Score: 662 / 750 (Mathematics: 141 / 150)

RESEARCH EXPERIENCE

First-author paper, ACL 2026 Main Conference (accepted) [OpenReview]
"DixitWorld: Evaluating Multimodal Abductive Reasoning in Vision-Language Models with Multi-Agent Dixit Gameplay"
Yunxiang Mo, Tianshi Zheng, Qing Zong, Jiayu Liu, Baixuan Xu, Yauwai Yim, Chunkit Chan, Jiaxin Bai, Yangqiu Song. Advisor: Prof. Yangqiu Song.

- Proposed DixitArena, a multi-agent environment that tests hypothesis generation (storyteller) and hypothesis selection (listener) under imperfect information, and DixitBench, a static QA benchmark (252 items over three difficulty tiers) that isolates the listener task
- Revealed a storyteller–listener asymmetry across six VLMs: over 78% of storyteller rounds score zero while the best listener reaches 75.6% accuracy; small open-source models are more creative storytellers, large proprietary models stronger listeners
- Validated DixitBench as a lightweight proxy for DixitArena ($r = 0.947$); a 72B-parameter scaling ablation and calibration/sensitivity analyses show the gap is structural; implemented in PyTorch with Transformer-based multimodal architectures

First-author paper, under review [arXiv]
"AgentIdeaBench: Benchmarking Scientific Ideation in the Agent Era"
Yunxiang Mo, Tianshi Zheng, Yisen Gao, Rui Wang, Newt Nguyen Kim Hue Nam, Kelvin Kiu Wai Tam, Jiaxin Bai, Yangqiu Song, Ginny Wong, Simon See.

- Built a multidisciplinary benchmark that evaluates scientific ideation under matched static-observation and active-exploration settings, scoring 33 LLMs across 40 subfields in five disciplines with literature-verified critics that judge originality against retrieved prior art
- Showed that active exploration exposes far more capability headroom: performance scales about twice as fast as under static observation, the gain is capability-gated toward the strongest models, and it comes from better grounding (feasibility, clarity, specificity) rather than originality
- Explored Scientific World Modeling, a generation-time loop that refines hypotheses through structured thought experiments; it helps mid-capability models but adds little for frontier models

Co-first-author paper, under review
"Robust Information-Gain Control for Active Agentic Reasoning under Approximate Beliefs"
Fang Wu*, Yunxiang Mo*, Aaron Tu, Zhenfei Yin, Shixiang Tang, Xiangru Tang, Tianshi Zheng, Yuheng Yang, Han Luo, Minghao Sun, Kejun Ying, Hejia Geng, Zehong Wang, Peng Xia, Yangqiu Song, Jure Leskovec. **Equal contribution.*

- Formalized Belief Trap Regions and showed that information-gain maximization locks LLM agents into self-reinforcing loops when their internal beliefs are only approximate
- Proposed Distributionally Robust Information Gain (DR-IG), a training-free decision-time objective that maximizes worst-case information gain around the current belief, with a verbalized belief proxy extracted from confidence-annotated outputs
- Across five LLM families, DR-IG cuts trap incidence to 0% at every tested shortlist size and recovers solve rates where IG-max solves nothing; in a real browser DOM it recovers a significant share of the 26.5% solve rate lost to belief approximation

Co-author paper, ICLR 2026 (Oral) [arXiv]
"ScaleCUA: Scaling Open-Source Computer Use Agents with Cross-Platform Data"
Zhaoyang Liu, Jingjing Xie, . . . , Yunxiang Mo, . . . , Wenhai Wang. *Contribution: data pipeline and cross-platform workflow components in the open-source codebase.*

INTERNSHIP EXPERIENCE

Stanford HAI, Stanford University Apr 2026 – Present
Research Intern Mentored by Dr. Fang Wu in the groups of Prof. Yejin Choi and Prof. Jure Leskovec

- Research on robust, belief-aware decision-making for LLM agents in multi-turn reasoning (co-first-author DR-IG paper above)

Beijing Ingenic Semiconductor Co., Ltd. Jun 2025 – Aug 2025
Machine Learning Engineer Intern

- Developed and optimized machine learning models for embedded and on-chip AI scenarios
- Participated in model training, evaluation, and inference pipeline development using PyTorch
- Assisted in deploying ML models to edge devices with constraints on latency and memory
- Collaborated with algorithm and hardware teams to align model design with chip-level constraints

Benchmark Architectural Design Co., Ltd Jan 2025

- Developed front-end modules using the MFC framework for a mini-program project
- Implemented UI design, event handling, and system debugging in a team environment

TECHNICAL SKILLS

Machine Learning & NLP: Transformer Architectures, Multimodal Reasoning, Abductive Inference, Prompt Engineering, Fine-tuning
Frameworks & Tools: PyTorch, Hugging Face Transformers, OpenAI API
Programming Languages: Python, C++, MATLAB
Languages: Chinese (Native), English (Fluent)

COMPETITIONS

- ACM-ICPC East Asia Regional Contest – Finalist / Honorable Mention
- National Mathematical Olympiad (China) – Second Prize, 2023
- National Informatics Competition (China) – Second Prize, 2022